

$$\int \sum \sum \pi \sqrt{x} \lim \sum \theta \Delta x$$

CALCULUS

WITH ASANTE

TOPIC ONE — ANSWER KEY | TOPIC SERIES

Series, Limits of a Series & p-Series

Full worked solutions for the Problem Set. Try every question yourself before reading these.

$$\int \sum \sum \pi \sqrt{x} \lim \sum \theta \Delta x$$

Compiled and taught by Asante

MATH 152 | Calculus & Analysis | KNUST BME1

Problem Set — Worked Solutions

1. $\sum \frac{1}{n^5}$

Correct answer: (C)

This is a p-series with $p = 5$. Since $p > 1$, the series **converges**.

2. $\sum \frac{1}{n^{0.5}}$

Correct answer: (A)

This is a p-series with $p = 0.5$. Since $p \leq 1$, the series **diverges**.

3. $\sum \frac{1}{n}$

Correct answer: (B)

The harmonic series **diverges**. The n -th term test only catches divergence when $a_n \not\rightarrow 0$ — here $a_n = 1/n \rightarrow 0$, so the test is silent. It never certifies convergence, so grouping terms (or the p-series test with $p = 1$) is needed to see the partial sums grow without bound.

4. For which p does $\sum \frac{1}{n^p}$ diverge?

Correct answer: (A)

$p \leq 1$. This includes the boundary case $p = 1$ (the harmonic series) and every smaller exponent.

5. $\lim_{n \rightarrow \infty} \frac{7n - 2}{4n + 9}$

Correct answer: (B)

Divide top and bottom by n :

$$\lim_{n \rightarrow \infty} \frac{7 - \frac{2}{n}}{4 + \frac{9}{n}} = \frac{7}{4}$$

6. $\lim_{n \rightarrow \infty} \frac{3n^2 + n}{n^3 + 1}$

Correct answer: (D)

Divide top and bottom by n^3 , the highest power present:

$$\lim_{n \rightarrow \infty} \frac{\frac{3}{n} + \frac{1}{n^2}}{1 + \frac{1}{n^3}} = \frac{0 + 0}{1 + 0} = 0$$

The denominator's degree (3) beats the numerator's (2), so the sequence is squeezed to 0.

7. $\sum \frac{n}{n+1}$ — **n -th term test**

Correct answer: (C)

$$\lim_{n \rightarrow \infty} \frac{n}{n+1} = \lim_{n \rightarrow \infty} \frac{1}{1 + \frac{1}{n}} = 1 \neq 0$$

Since the terms don't approach 0, the series **diverges** by the n -th term test — no partial-sum work required.

8. $\sum \frac{1}{n^{1.01}}$

Correct answer: (B)

$p = 1.01 > 1$, so the series **converges** — but only barely. It sits just above the boundary case $p = 1$ (the harmonic series, which diverges). This is the sharpest possible illustration that convergence is decided by a strict inequality: $p = 1.01$ converges, $p = 0.99$ diverges, and there's no partial credit near the line, however slowly the series grows.